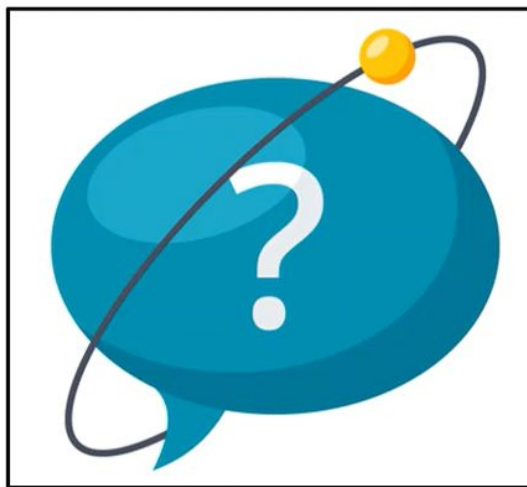


w4.1- One Sample Tests (7:36)

w4.1- One Sample Tests (7:36).mp4

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Module 4 - Lesson 1



The nonparametric tests are often
called distribution free,



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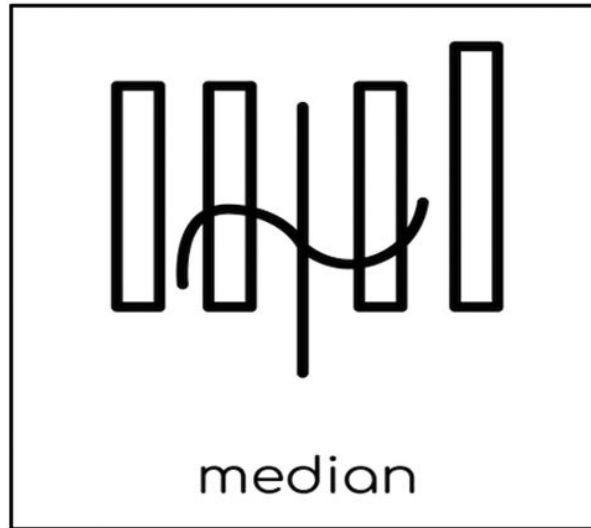
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Mood's Median Test



00:01:15 / 00:07:36

Mood's Median Test Example

A	B	C
32	15	61
20	23	53
21	50	12
43	33	28
48	41	36
52		44
17		

Overall median = 34.5

	A	B	C	Total
Greater than Median	3	2	4	9
Less than or equal to Median	4	3	2	9
Total	7	5	6	18

Test statistic: $\chi^2_{test} = 1.0095$

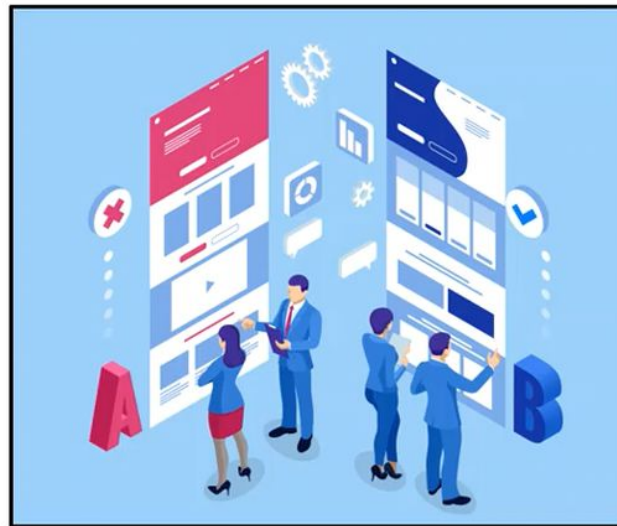
Critical statistic: $\chi^2_{crit} = 5.991$

Fail to reject the null



Let's consider this example.

Levene's Test



Levine's test is the go to procedure for determining homogeneity of variances.



Levene's Test Example

	A	B	C
	32	15	61
	20	23	53
	21	50	12
	43	33	28
	48	41	36
	52		44
	17		
Mean	33.28571	32.4	39

Column	n	Mean	Std. Dev.	Std. Error
A	7	12.33	5.654	2.137
B	5	10.72	7.082	3.167
C	6	13.67	9.416	3.844

Absolute Value of Residuals		
A	B	C
1.285714	17.4	22
13.28571	9.4	14
12.28571	17.6	27
9.714286	0.6	11
14.71429	8.6	3
18.71429		5
16.28571		

Source	DF	SS	MS	F-Stat	P-value
Columns	2	23.680487	11.840244	0.21250394	0.8109
Error	15	835.76641	55.71776		
Total	17	859.44689			

Fail to reject the null



Again, assume alpha is 0.05.

Sign Test

- Applications
 - Matched pairs
 - Nominal majority
 - Median verses target
- Test statistic:
 - If $n \leq 25$, use the smaller number of + or – signs.
 - If $n > 25$, use the smaller number of + or – signs and convert to a z score

$$z = \frac{(x + 0.5) - \left(\frac{n}{2}\right)}{\frac{\sqrt{n}}{2}}$$



Sign Test Example

- Heights of males
- 117 subjects
- 71 had heights below 76 inches
- 37 had heights above 76 inches
- 9 had heights equal to 76 inches

$$Z = \frac{(x+0.5) - \frac{n}{2}}{\frac{\sqrt{n}}{2}} = \frac{(37+0.5) - \frac{108}{2}}{\frac{\sqrt{108}}{2}} = -3.17543$$

p-value = 0.00074807

Reject the null



This module from the Six Sigma Black Belt course introduces **non-parametric hypothesis testing**. These methods are essential when data **does not meet the strict assumptions** (like normality) required by parametric tests.

1. Overview of Non-Parametric Tests

Non-parametric tests are often called "**distribution-free**" because they do not assume the population follows a specific distribution (e.g., a normal curve).

- **When to use:** When assumptions are violated, when data is ranked/ordinal, or when a more conservative alternative is needed.
- **Advantages:** Greater flexibility; can be applied to more data types.
- **Disadvantages:** Less efficient than parametric tests; they "waste" information and may require larger sample sizes to achieve the same power.

2. Mood's Median Test

Used to compare the **medians** of two or more independent groups to see if they differ significantly.

- **Mechanism:** It is a variation of the Chi-Square contingency table test. It counts how many values in each group fall above or below the overall median of all combined data.
- **Hypotheses:**
 - H_0 : All group medians are equal.
 - H_a : At least one group median is different.
- **Example Outcome:** If the calculated Chi-Square statistic is less than the critical value, you fail to reject the null hypothesis (medians are the same).

3. Levene's Test

Levene's test is the primary tool for checking **homogeneity of variances** (whether groups have equal spread).

- **Robustness:** While technically parametric, it is highly robust against violations of normality, making it a "go-to" procedure.
- **Process:**
 1. Calculate the **absolute residuals** (the difference between each data point and its group mean).
 2. Run an **ANOVA** on those absolute residuals.
- **Decision Rule:** If Levene's test shows significant differences in variance ($p < 0.05$), researchers should switch to **Welch's t-test**, which does not assume equal variances.

4. The Sign Test

The most basic non-parametric test, used for matched pairs or comparing a median to a target value.

- **Mechanism:** Converts data into **plus (+)** and **minus (-)** signs based on whether they are above or below a target. It ignores "ties."
- **Test Statistic (X):**
 - **Small Samples ($n \leq 25$):** Uses the frequency of the less common sign.
 - **Large Samples ($n > 25$):** Converts the frequency into a **Z-score** using a specific formula.
- **Example Case:** Testing if a median height is less than 76 inches. If the resulting p-value is very low (e.g., 0.0007), the null hypothesis is rejected in favor of the alternative.

Comparison Summary

Test	Purpose	Parametric Equivalent
Mood's Median	Compare medians of 2+ groups	ANOVA
Levene's Test	Check equality of variances	F-Test (less robust)
Sign Test	Compare median to target / Matched pairs	One-sample t-test / Paired t-test

for the non parametric test should we rank sort our sample first

Yes, for the vast majority of non-parametric tests, **ranking your data** is the fundamental first step.

Since these tests are "distribution-free," they don't care about the specific distance between values (the interval); they only care about the **order** of the values. By converting raw data into ranks, you neutralize the impact of outliers and skewed distributions.

w4.2- Two Sample Tests (8:01)

Wilcoxon Signed Ranked Test

- Applications
 - Matched pairs comparison of medians
 - Comparing a median to a target value
- Assumptions
 - Simple random sample
 - Distribution of the differences is approximately symmetric



The Wilcoxon signed-rank test is

Wilcoxon Signed Ranked Test

- Test statistic
 - If $n \leq 30$, Use the smaller of
 - Sum of the positive ranks of nonzero differences
 - Absolute value of the sum of the negative ranks of nonzero differences
 - If $n > 30$, use this statistic, T calculated per above

$$Z = \frac{T - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}$$



To find the test statistic,

Wilcoxon Signed Ranked Test Example

Football	Soccer
25	60
53	42
19	31
32	29
40	39
42	51
26	55
23	31
27	24
12	10

Variable	n	n for test	Median Est.	Wilcoxon Stat.	P-value
var1-var2	10	10	-4.5	17	0.3077

Fail to reject the null



Consider this example.

Mann-Whitney

- Applications
 - Comparing the medians of two independent samples
- Assumptions
 - Simple random sample
 - Sample size of each group is 10 or more

sample data from two
independent populations



Mann-Whitney

- Test statistic: $z = \frac{R - \mu_R}{\sigma_R}$

- Where $\mu_R = \frac{n_1(n_1 + n_2 + 1)}{2}$ $\sigma_R = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$

n_1 = size of the sample from which the rank sum R is found

n_2 = size of the other sample

R = sum of ranks of the sample with size n_1



It is a standardized sum of ranks

Mann-Whitney Example

Control	Treatment
5	20
11	19
3	21
2	18
22	15
10	6
13	12
9	16
6	9
7	13
13	
20	

Test statistic: $Z = \pm 1.78033$

P-value = $0.03722 \times 2 = 0.07445$

Fail to reject the null

$$Z = \frac{\text{Min } U - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}}$$



Let's consider this example.

Spearman's Ranked Correlation

- Applications
 - Assessing the association between two paired variables
- Assumptions
 - Simple random sample
 - Data are ranks or can be converted to ranks
- Advantages
 - Does not require a normal distribution
 - Detect some non-linear relationships



Spearman's Ranked Correlation

- Critical value
 - If $n \leq 30$, use the critical correlation value
 - If $n > 30$, $r_s = \frac{\pm z}{\sqrt{n-1}}$



Spearman's Ranked Correlation

- Test statistic (general form):

$$r_s = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \sqrt{n(\sum y^2) - (\sum y)^2}}$$

- Test statistic (no ties):

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Take note that the data
for the x s and y s must be



Spearman Example

x	y
10	3
8	10.5
6	11
2	15
13	6
12.5	4
6	2.5
9	12
13	5
8	8
5	6
6	14
2	12
15	1.5
4	10.5

Sample correlation: -0.63874289

P-value = 0.0104

Reject the null



This module continues the exploration of non-parametric tests, specifically focusing on comparing **two groups** and testing for **correlation**. These tests are invaluable when data is not normally distributed or consists of ranks.

1. Wilcoxon Signed-Rank Test

This test is used to compare the paired differences of medians between two groups or to compare a median to a target value.

- **Parametric Equivalent:** Paired t-test.
- **Comparison to Sign Test:** Unlike the Sign Test (which only looks at whether a difference is positive or negative), the Wilcoxon test considers the **magnitude** of the differences, making it more powerful.
- **Key Assumption:** The distribution of the differences should be roughly bell-shaped.
- **Test Statistic (T):**
 - $n \leq 30$: The smaller of the sum of positive or negative ranks.
 - $n > 30$: The T value is converted to a **Z-score**.

2. Mann-Whitney Test

Used to test the equality of medians from **two independent populations**.

- **Parametric Equivalent:** Independent samples t-test.
- **Hypotheses:**
 - H_0 : Population medians are the same.
 - H_a : Population medians are different (or one is specifically greater/less).
- **Test Statistic:** Often reported as a **U statistic**. When sample sizes are large ($n > 20$), the U distribution approximates a normal distribution, allowing for conversion to a **Z-score**.
- **Requirements:** Samples must be simple random samples with at least 10 values each.

3. Spearman's Rank Correlation (ρ)

Used to test for an association or correlation between two variables using matched pairs.

- **Parametric Equivalent:** Pearson Correlation (r).
- **Advantages:** Does not require normality and is effective at identifying **non-linear** (monotonic) relationships.
- **Process:** 1. Convert raw data (x and y) into **ranks**. 2. Calculate the correlation based on these ranks.
- **Decision Rule:** Compare the calculated Rho (ρ) or the resulting p-value to alpha (α). If $p < \alpha$, a significant correlation exists.

Two-Group Test Summary

Test	Data Relationship	Purpose	Equivalent Parametric Test
Wilcoxon Signed-Rank	Paired / Matched	Compare paired medians	Paired t-test
Mann-Whitney	Independent	Compare independent medians	Independent t-test
Spearman's Rho	Paired / Matched	Test for association	Pearson Correlation

w4.3- Multi-Sample Tests (4:43)

Kruskal Wallis

- Applications
 - Comparing the medians of three or more independent samples
- Assumptions
 - Simple random sample
 - Sample size of each group is 5 or more
- Test statistic:

$$H = \frac{12}{N(N+1)} \left(\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \dots + \frac{R_k^2}{n_k} \right) - 3(N+1)$$



Kruskal Wallis Example

Control	Treatment 1	Treatment 2
5	20	34
11	19	59
3	21	23
2	18	29
22	15	37
10	6	16
13	12	62
9	16	
6	9	
7	13	
13		
20		

Test statistic: $H = 15.043$

P-value = 0.00054

Reject the null

Let's look at an extension
of the last example.



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Friedman Test

- Applications
 - Comparing the medians of three or more paired samples (treatments)
- Assumptions
 - Simple random sample
 - No interaction between rows and treatment levels

The Friedman test is a nonparametric test that uses ranks to compare



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Friedman Test

- Test statistic:

$$Q = \frac{12}{nk(k+1)} \sum_{j=1}^k R_j^2 - 3n(k+1)$$

- R_j = sum of ranks for sample j
- n = number of rows
- k = number of treatments

$R_{\text{sub } j}$ is the sum of the ranks for
sample j ,



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Friedman Test Example

Person	Fizz	Diet Fizz	Fizz Zero
1	9	7	8
2	7	4	4
3	6	8	6
4	10	8	3
5	5	6	6
6	3	7	4
7	6	9	2
8	7	5	8
9	6	3	4
10	8	6	5
11	3	8	2
12	8	4	7

Test statistic: $Q = 2.0417$

P-value = 0.36029



Let's consider this example.

This module covers non-parametric tests used when comparing **three or more groups**. These are powerful alternatives to ANOVA when your data is small, non-normal, or contains outliers.

You use **ANOVA (Analysis of Variance)** when you want to determine if there is a statistically significant difference between the **means of three or more independent groups**. 

While a t-test is great for comparing two groups, using multiple t-tests to compare three or more groups increases the risk of a **Type I error** (false positive). ANOVA solves this by testing all groups simultaneously.

1. Kruskal-Wallis Test

The Kruskal-Wallis test is used to compare the medians of three or more **independent** groups.

- **Parametric Equivalent:** One-way ANOVA.
- **Best Use Case:** When samples are small, unequal in size, or deviate strongly from a normal distribution.
- **Mechanism:** It ranks all data points from all groups combined and then calculates a test statistic (H) based on the sum of the ranks for each group (R_k).
- **Hypotheses:**
 - H_0 : All population medians are equal.
 - H_a : At least one population median is different.
- **Distribution:** The test statistic H follows a **Chi-Square distribution** with $k - 1$ degrees of freedom (where k is the number of groups).

2. Friedman Test

The Friedman test is the non-parametric choice for comparing three or more **dependent** (paired or repeated) groups.

- **Parametric Equivalent:** Repeated measures ANOVA.
- **Example Scenario:** A "taste test" where the same group of people rates three different **soda brands**. Because the same person is testing all three, the data is "blocked" or paired by person.
- **Assumptions:**
 - Simple random samples.
 - No interaction between the rows (blocks) and the treatment levels.
- **Test Statistic (Q):** Similar to Kruskal-Wallis, it uses ranks **within each block** and follows a **Chi-Square distribution** with $k - 1$ degrees of freedom.

Comparison of Multi-Group Tests

Test	Group Relationship	Data Requirement	Parametric Equivalent
Kruskal-Wallis	Independent	≥ 5 values per sample	One-way ANOVA
Friedman	Dependent (Paired)	Matched sets/Repeated measures	Repeated Measures ANOVA

Decision Rule Reminder

For both tests, the decision process remains the same:

1. If $P\text{-value} < \alpha$: Reject the null hypothesis. There is a significant difference between the medians.
2. If $P\text{-value} \geq \alpha$: Fail to reject the null. There is no significant evidence of a difference.